

SliceGAN with Vision Transformer discriminators and a SAM front-end

Vision Transformers — FIUBA.

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1. Project goal

Three-dimensional microstructures are needed to compute effective material properties (for example as representative volume elements for finite-element analysis), but 3D imaging (micro-CT, FIB-SEM) is expensive and often unavailable, while 2D micrographs are cheap, fast and usually of higher resolution. This project generates 3D two-phase microstructures from a single 2D micrograph with SliceGAN (Kench & Cooper, 2021) and studies whether Vision Transformers improve it, either as the adversarial critic or as a segmentation front-end.

1.1 Background: SliceGAN

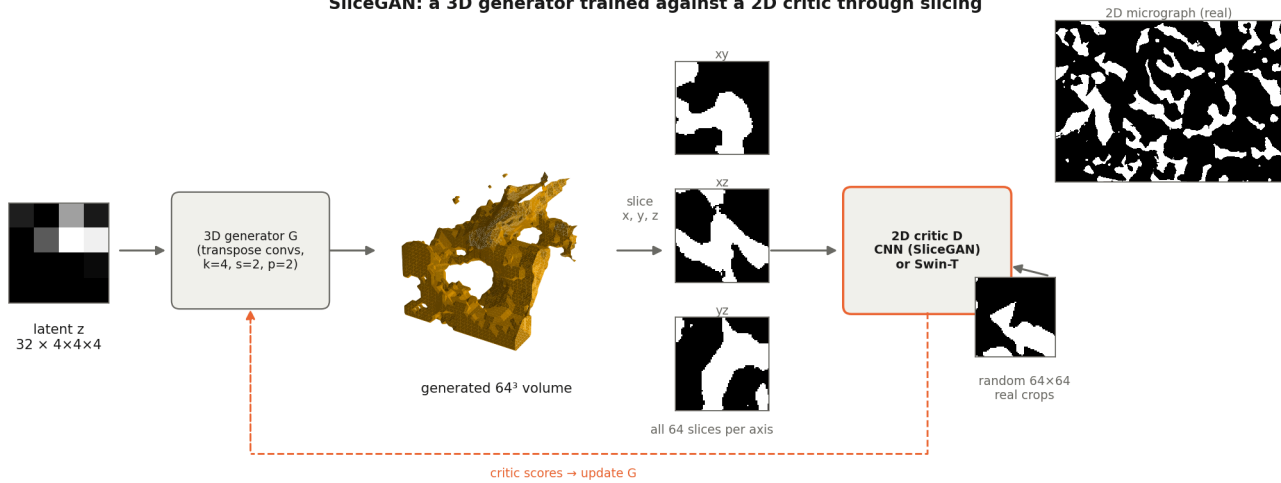
The problem. A 2D micrograph of a material contains, statistically, much of the information of its 3D structure: for an isotropic material (no preferred direction), every planar cut through the volume has the same statistics (phase fractions, feature sizes and shapes, spatial correlations) as any other. The task is therefore to produce 3D volumes whose 2D sections are indistinguishable, statistically, from the micrograph. Classical reconstruction methods optimize a 3D volume to match chosen statistical descriptors (e.g. the two-point correlation), and they are slow (hours for 10^6 voxels) and only reproduce the descriptors they were told to match.

Generative adversarial networks. A GAN trains two networks against each other: a *generator* G that maps random noise to samples, and a *discriminator* (critic) D that tries to tell generated samples from real ones. G is updated to fool D . At equilibrium the generated distribution matches the real one. GANs learn the statistics directly from the data instead of from hand-picked descriptors, and once trained they generate new samples in seconds.

SliceGAN's key idea: dimensionality expansion through slicing (Kench & Cooper, 2021). A GAN normally needs training data of the same dimensionality as its output, but no 3D training data exists here. SliceGAN combines a **3D generator** with a **2D discriminator** and bridges them with a slicing step:

1. G maps a latent tensor z (Gaussian noise of shape $32 \times 4 \times 4 \times 4$ in this project) to a 64^3 volume with one channel per phase (softmax, i.e. a soft one-hot encoding).
2. The generated volume is cut into all 64 slices along each of the three axes ($3 \times 64 = 192$ slices).
3. D , a 2D convolutional network, scores each slice and an equal-sized random crop of the real micrograph. For an isotropic material one D serves all three directions.
4. The losses of all slices are combined, and both networks are updated. Because every slice of the volume must look like the micrograph, G learns a 3D structure whose sections in x , y and z all match the 2D statistics.

SliceGAN: a 3D generator trained against a 2D critic through slicing



Wasserstein loss over all slices: D learns to separate real crops from generated slices; G is updated so that every slice looks real

SliceGAN training

Training in the original version uses the Wasserstein loss with gradient penalty (WGAN-GP; Gulrajani et al., 2017), in which D is an unbounded *critic* estimating the Wasserstein distance between real and generated slices, with 5 critic updates per generator update. Using a WGAN means that this distance is only valid if the critic is a 1-Lipschitz function (gradient norm with respect to the input ≤ 1). Instead of clipping the weights as the original WGAN does, the GP enforces the constraint softly: a random interpolation between a real slice x and a generated one $\tilde{x}$, $\hat{x} = \epsilon x + (1-\epsilon)\tilde{x}$ with $\epsilon \sim U[0, 1]$, is taken, and the term $\lambda(\|\nabla D(\hat{x})\|_2 - 1)^2$, with $\lambda = 10$, is added to the critic loss. The discriminator sees all 64 slices per direction of every generated volume and uses a generator batch twice the critic batch ($m_G = 2 m_D$), which the authors found most efficient.

Generator design: uniform information density. Early SliceGAN versions produced worse quality near volume edges. The cause is transpose convolution: a voxel near the edge of the output receives contributions from fewer kernel positions than a central voxel, so information is unevenly distributed. For microstructures, where edges matter as much as the centre, the authors derive rules for the kernel size k , stride s and padding p ($s < k$, $k \bmod s = 0$, $p \geq k - s$) and use $\{k, s, p\} = \{4, 2, 2\}$. They also give the latent z a spatial size of 4 instead of 1, so that the first layer already learns overlapping kernel outputs. As a consequence, volumes larger than 64^3 can be generated after training by simply enlarging z . The critic, in turn, is a plain 2D CNN of five strided convolutions (64×64 slice $\rightarrow$ one score).

Scope and limits. SliceGAN reproduces the micrograph's statistics without hand-picked descriptors, trains in a few hours on one GPU and generates volumes in seconds. It was validated against real 3D data of a battery electrode and later applied to 87 materials in the MicroLib library (Kench et al., 2022). In this work it is restricted to isotropic cases (anisotropic materials need two or three perpendicular micrographs and separate critics) and a field of view that is representative of the material.

1.2 Background: Swin-T and SAM, and where they enter the pipeline

Swin Transformer (Swin-T; Liu et al., 2021) is a hierarchical Vision Transformer:

- the image is split into 4×4 patches, each embedded as a token;
- self-attention is computed inside local 7×7 -token windows, which shift between consecutive layers so information also flows across windows;
- four stages merge patches between them, giving feature maps at decreasing resolution (like a CNN);
- position is encoded by a relative-position bias inside each window, not by absolute embeddings, so the network also runs on small 64×64 slices;
- Swin-T has 28 M parameters and is pretrained on ImageNet-1k.

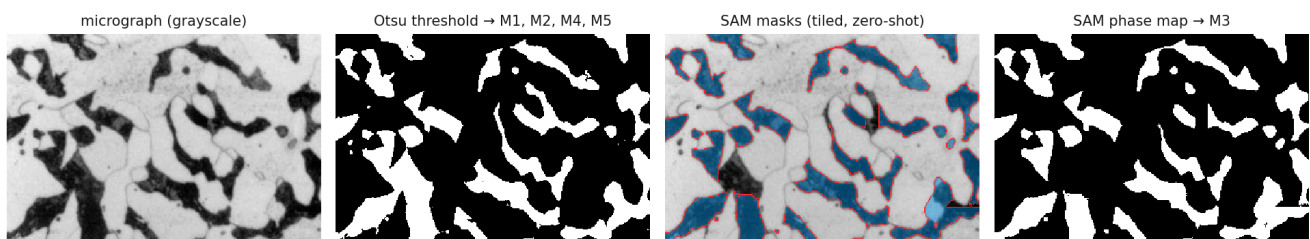
In this project Swin-T replaces (M2, M3) or complements (M4, M5) SliceGAN’s CNN as the 2D critic, bringing pretrained features and attention over the whole slice. Transformer critics are known to destabilize GAN training (ViTGAN; Lee et al., 2022), which turned out to be the central difficulty (Section 5.2).

Segment Anything (SAM; Kirillov et al., 2023) is a promptable segmentation model: a ViT image encoder, a prompt encoder (points, boxes) and a light mask decoder, trained on 1 billion masks. In automatic mode a grid of point prompts yields a mask for every object, zero-shot.

How SAM is included. SliceGAN needs a segmented micrograph (one label per phase) as training data. The baseline approach obtains it with a global gray-level threshold computed with Otsu’s method. This works well for two-phase cases, and it can be extended to multi-phase ones. In this work, SAM is used as an alternative front-end for this segmentation step, and the rest of the pipeline is unchanged:

1. SAM ViT-B (facebook/sam-vit-base, no fine-tuning) runs on overlapping 256 px tiles of the micrograph with a 32×32 point grid per tile (a single pass over the whole image misses most small islands);
2. masks that overlap (IoU > 0.3) are merged;
3. each merged group is labelled inclusion or matrix by its mean gray level (a two-class split of the group means; uncovered pixels are matrix);
4. the resulting phase map replaces the Otsu map as the training image of model M3.

Phase segmentation of MicroLib 000210 (crop): Otsu ($\varphi = 0.232$) vs SAM ($\varphi = 0.217$)



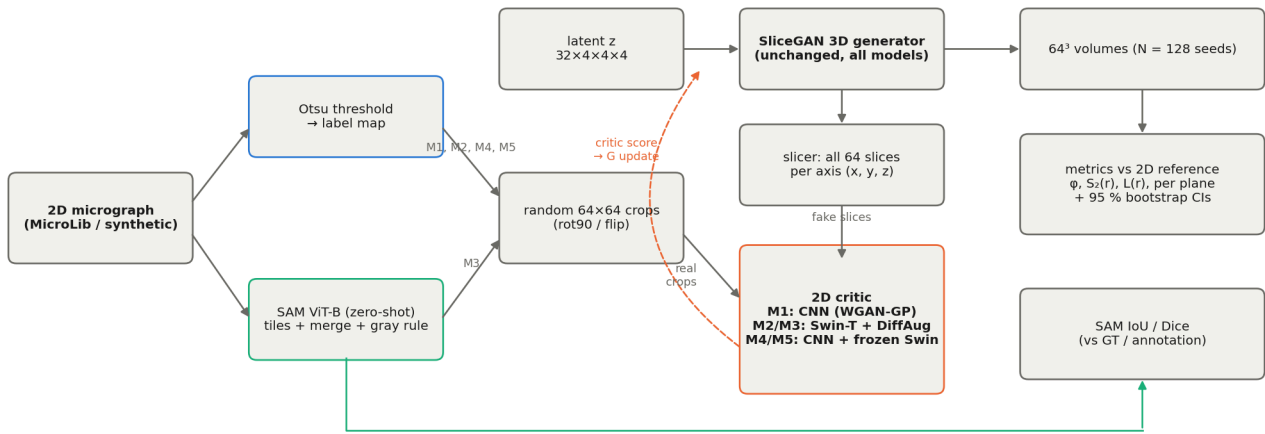
SAM front-end

1.3 Project questions

- **RQ1 — ViT critic.** Does replacing SliceGAN’s CNN discriminator by a Swin Transformer (Swin-T) improve the generated microstructures? (M1 vs M2, with an M1 + DiffAug ablation)
- **RQ2 — SAM front-end.** Does segmenting the micrograph with SAM, instead of a global threshold, improve them? (M2 vs M3)
- **RQ3 — ViT as an additional critic.** Does adding a pretrained Swin-T critic next to SliceGAN’s CNN, as in Vision-aided GAN (Kumari et al., 2022), help, either from scratch (M4) or as a fine-tuning stage of a trained SliceGAN (M5, compared against M1 trained for the same extra steps)?

Quality is measured with the statistical descriptors used in the microstructure-generation literature (phase fraction, two-point correlation, lineal path), since no 3D ground truth exists.

2. Overall architecture



Pipeline

Data. In the isotropic case the training set is a single 2D micrograph, and the same image serves as the reference for the three planes (xy, xz, yz). For anisotropic materials up to three images can be given, one per orthogonal plane, and each direction has its own critic.

In this work, the main case is MicroLib entry 000210 (DoITPoMS micrograph library): an optical micrograph of dark islands in a light matrix, 437×800 px after removing the scale bar, $0.687 \mu\text{m}/\text{px}$. It was chosen at random among the MicroLib two-phase entries with a phase gray-level gap ≥ 80 and accepted after an isotropy check (x/y correlation-length ratio 1.11).

The second case study is a synthetic micrograph (non-overlapping discs, $\phi = 0.25$, rendered with blur and noise, exact ground-truth mask). Unlike the previous case, the exact phase mask is known, so both the segmentation (Otsu or SAM) and the reference descriptors (ϕ , S_2) are measured without annotation error.

Segmentation (front-end). M1, M2, M4 and M5 train on an Otsu label map ($\phi = 0.232$). M3 trains on a SAM label map: zero-shot SAM ViT-B on 256 px tiles, overlapping masks merged (IoU > 0.3), and groups whose mean gray level departs from the matrix labelled as inclusions ($\phi = 0.217$).

Generator. SliceGAN’s 3D resize-convolution generator, unchanged for every model (40.1 M parameters): a $32 \times 4 \times 4 \times 4$ Gaussian latent $\rightarrow 64^3$ volume, softmax over the two phases.

Slicer and critics. Each generated volume is cut into all 64 slices along x, y and z. The slices and real 64×64 crops of the training image are scored by a 2D critic:

Model	Training image	2D critic	Loss	Trainable critic params
M1	Otsu	SliceGAN CNN (5 strided convs)	WGAN-GP	2.8 M
M2	Otsu	Swin-T (ImageNet), stages 3–4 + head trainable, DiffAug, lr 2e-5	WGAN-GP	26.3 M
M3	SAM	same critic as M2	same as M2	same as M2
M4	Otsu	SliceGAN CNN + frozen Swin-T with per-scale heads (ensemble)	CNN: WGAN-GP, Swin: hinge	2.8 M + 0.18 M
M5	Otsu	as M4, generator initialised from M1’s best checkpoint	as M4	as M4
M1-ext	Otsu	as M1, generator initialised from M1’s best checkpoint	WGAN-GP	2.8 M

Metrics. For each model, 128 64^3 volumes (seeds 0–127) are compared with the 2D image through ϕ , $S_2(r)$ and $L(r)$, overall and per slice orientation, with bootstrap confidence intervals (Section 4).

3. Technical implementation

Stack. Python 3.11, PyTorch 2.5.1 (CUDA 12.4), timm 1.0.11, Hugging Face transformers 4.46.3, NumPy/SciPy/scikit-image, Poetry environment (setup.sh), YAML configs with inheritance and dataset overlays, logging to file and console, MLflow experiment tracking, 92 pytest tests. Trained on one NVIDIA RTX A2000 (12 GB).

Pretrained models.

- Swin-T: timm/swin_tiny_patch4_window7_224.ms_in1k on the Hugging Face Hub (the Microsoft ImageNet-1k checkpoint, same weights as microsoft/swin-tiny-patch4-window7-224). timm is used instead of transformers because Swin-T runs natively on 64 px slices only if the attention windows of stages 3–4 shrink to 4×4 and 2×2 with a resized relative-position bias, which timm supports and transformers 4.46 does not. The RGB patch embedding is converted to two phase channels so that a one-hot slice is seen as a centred gray image (matrix -0.5 , inclusion $+0.5$).
- SAM: facebook/sam-vit-base through the transformers mask-generation pipeline, zero-shot.

SliceGAN integration. The upstream repository is a git submodule (external/SliceGAN, MIT) and is not modified.

- src/models/slicegan_wrapper.py builds the generator and CNN critic with upstream’s network factory and the exact layer lists of run_slicegan.py, and reuses its gradient penalty.
- src/models/train.py is a fork of upstream’s training loop with the same WGAN-GP schedule (Adam $1e-4$, $\beta = (0.9, 0.99)$, $\lambda_{GP} = 10$, 5 critic steps per generator step) and SliceGAN’s batch rule: the critic sees all

64 slices per axis of $m_D = 1$ volume, the generator step uses $m_G = 2 m_D$ volumes.

- Additions: critic branches (single critic or CNN + Swin ensemble), hinge loss, differentiable augmentation of critic inputs (DiffAug: translation, cutout, 90° rotations/flips), generator warm start, and checkpoint selection.
- Translations (up to ± 8 px) and cutout positions are deliberately not aligned to Swin-T's 4 px patch grid: the real crops are taken at arbitrary pixel offsets too, and unaligned shifts change the content of every patch, so the critic cannot exploit where a feature falls within a patch.

Checkpoint selection and evaluation protocol. After every epoch (100 generator steps) the generator produces 16 volumes from held-out seeds (1000–1015), and the epoch with the lowest S_2 MAE against the model's own training image is kept. The kept checkpoint is then evaluated on 128 *different* seeds (0–127), so the reported numbers are never measured on the volumes used to choose the checkpoint. The same rule is applied to every model. The protocol was tightened twice during the project, because single 64^3 volumes are small samples of the microstructure (≈ 30 islands; phase fraction varies by ± 0.05 from volume to volume): with 2 selection seeds and 4 evaluation seeds (run v5), model rankings reversed when re-evaluated on 32 seeds, and a checkpoint picked as best on 2 seeds could be three times worse than the last epoch on fresh seeds. Even with 16 seeds, choosing the best of 50 noisy epoch scores is optimistic (the winner's curse: M1's selected checkpoint scored S_2 MAE 0.0016 on its selection seeds but 0.020 on 128 fresh seeds); evaluation on independent seeds removes that bias from the reported numbers.

RVE export. `src/models/export_rve.py` writes generated microstructures for FEM/FFT homogenization codes (VTK `.vti`, MetaImage `.mhd/.raw`, Abaqus `.inp` voxel mesh with phase element sets and face node sets, NumPy, TIFF, plus JSON metadata). Volumes larger than 64^3 are obtained by enlarging the latent input; periodic RVEs use SliceGAN's latent tiling with a 2-voxel crop, which we found to be the correct period (the upstream 1-voxel crop duplicates a slice at the seam); periodicity is verified by comparing the wrap-around face mismatch with interior slice-to-slice mismatch (ratio 1.0–2.0 vs 8–13 without tiling).

Main modules.

- `src/data/make_dataset.py`: download, crop, Otsu, crops.
- `src/features/sam_segment.py`: SAM front-end.
- `src/features/descriptors.py`: ϕ , S_2 , L .
- `src/models/discriminator_swin.py`: Swin critics.
- `src/models/train.py`.
- `src/models/generate.py`: volumes + metrics.
- `src/visualization/visualize.py`: figures, `metrics.csv`.
- `reports/viewer/`: an interactive volume viewer.
- `notebooks/00_eda.ipynb`: EDA notebook.

4. Evaluation

There is no 3D ground truth (no micro-CT), so generated volumes are compared statistically with the 2D training image. For an isotropic material, any planar section of the 3D microstructure has the same statistics as the 2D micrograph, so the 2D descriptors of the training image must be matched by the slice-averaged descriptors of the generated volume; this is the justification for the metrics below (SliceGAN, Kench & Cooper 2021). All are implemented in `src/features/descriptors.py` and tested in `tests/test_descriptors.py` against cases with analytic answers. Label 1 is the inclusion/pore phase and $I(x)$ its indicator function; curves are evaluated for $r = 0 \dots 32$ px on $N = 128$ generated 64^3 volumes per model (different seeds; the brief requires $N \geq 4$).

- **Phase fraction** $\phi = \langle I(x) \rangle$, reported as mean $\pm$ std over the N volumes, together with $|\Delta\phi| = |\phi_{\text{gen}} - \phi_{\text{train}}|$.
- **Two-point correlation** $S_2(r) = P[I(x) = 1, I(x + r) = 1]$, computed by FFT autocorrelation (each lag normalized by its number of valid pixel pairs, no periodicity assumed) and radially averaged over lag vectors with $\text{round}(|r|) = r$. $S_2(0) = \phi$ and $S_2(r) \rightarrow \phi^2$ for uncorrelated points. For a volume, S_2 is averaged over all xy , xz and yz slices (192 slices for 64^3).
- **Lineal path** $L(r)$ (as in Micro3Diff, Lee & Yun 2024; Lyu & Ren 2024): probability that a straight segment of $r + 1$ pixels lies entirely in the inclusion phase, averaged over both in-plane directions (2D) or the three axes (3D). $L(0) = \phi$; unlike S_2 , L is sensitive to connectivity along lines.
- **Errors**: MAE over r of the generated vs training curves (S_2 MAE, L MAE, using the mean curve over the N volumes), and the Micro3Diff error rate $\text{err}(S_2) = \text{mean}|S_{2,\text{gen}} - S_{2,\text{train}}| / \text{mean}|S_{2,\text{train}}|$ (likewise $\text{err}(L)$).
- **3D isotropy check**: ϕ , S_2 MAE and L MAE per slice orientation (xy , xz , yz). If one plane diverges, the volume is not isotropic. Averaged over all slices, $\phi_{xy} = \phi_{xz} = \phi_{yz} = \phi$ by construction, so the spread of the per-slice ϕ along each axis is reported too.
- **Uncertainty**. 95 % confidence intervals for $|\Delta\phi|$, S_2 MAE and L MAE come from a bootstrap over the 128 volumes (2000 resamples, each recomputing the metric exactly as reported). Each model is compared with the baseline M1 through the bootstrap distribution of the difference; a model is called better or worse only if that interval excludes zero.
- **Typical quality during training (selection-free)**. The per-epoch held-out scores are unbiased individually; only picking their minimum is optimistic. The median and interquartile range of the per-epoch held-out S_2 MAE over the second half of training, and the number of collapsed epochs (held-out $\phi < 0.05$), summarize how good and how stable a model is without any selection.
- **Common reference**. Every model is scored against its own training image and against a reference shared by all models of a dataset, so that M3 (trained on the SAM map) is comparable with M2: the Otsu map for MicroLib (no ground truth exists) and the exact ground-truth mask for synthetic data.
- **SAM segmentation quality** (M3 front-end): $\text{IoU} = |P \cap G| / |P \cup G|$ and $\text{Dice} = 2|P \cap G| / (|P| + |G|)$ of the label map P against the exact ground truth G on the synthetic image, and, on MicroLib, against a curated reference: the midpoint of the two phase gray levels annotated by the MicroLib authors (8 and 133) applied to the raw micrograph.

Reference scales: on the synthetic dataset, the Otsu segmentation vs the ground-truth mask gives S_2 MAE = 0.0021. Uncorrelated random 64^3 volumes with the correct ϕ of MicroLib 000210 give S_2 MAE = 0.042 ($\text{err} = 0.42$) and L MAE = 0.077 ($\text{err} = 0.89$), a floor any useful model must beat.

5. Results and examples

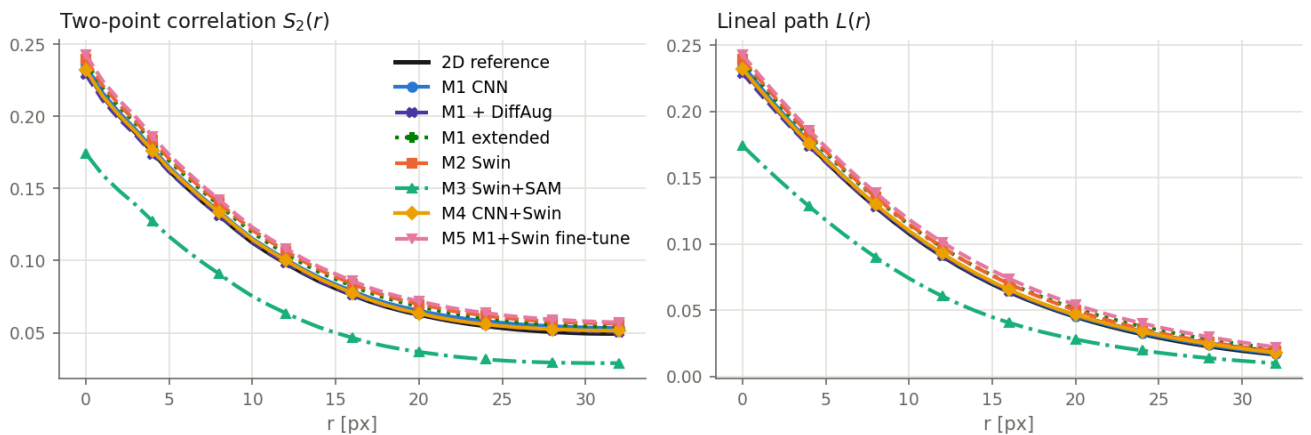
5.1 Main results (MicroLib 000210)

Train / validation / test protocol (Section 3): each model's checkpoint is chosen among the per-epoch best, the last epoch and the training snapshots on 128 validation seeds, then evaluated on 128 test volumes against the common reference (Otsu map, $\phi = 0.232$). Brackets: 95 % bootstrap intervals. "Typical" is the selection-free median held-out S_2 MAE over the second half of training (16 seeds per epoch, so on a different scale from the test columns).

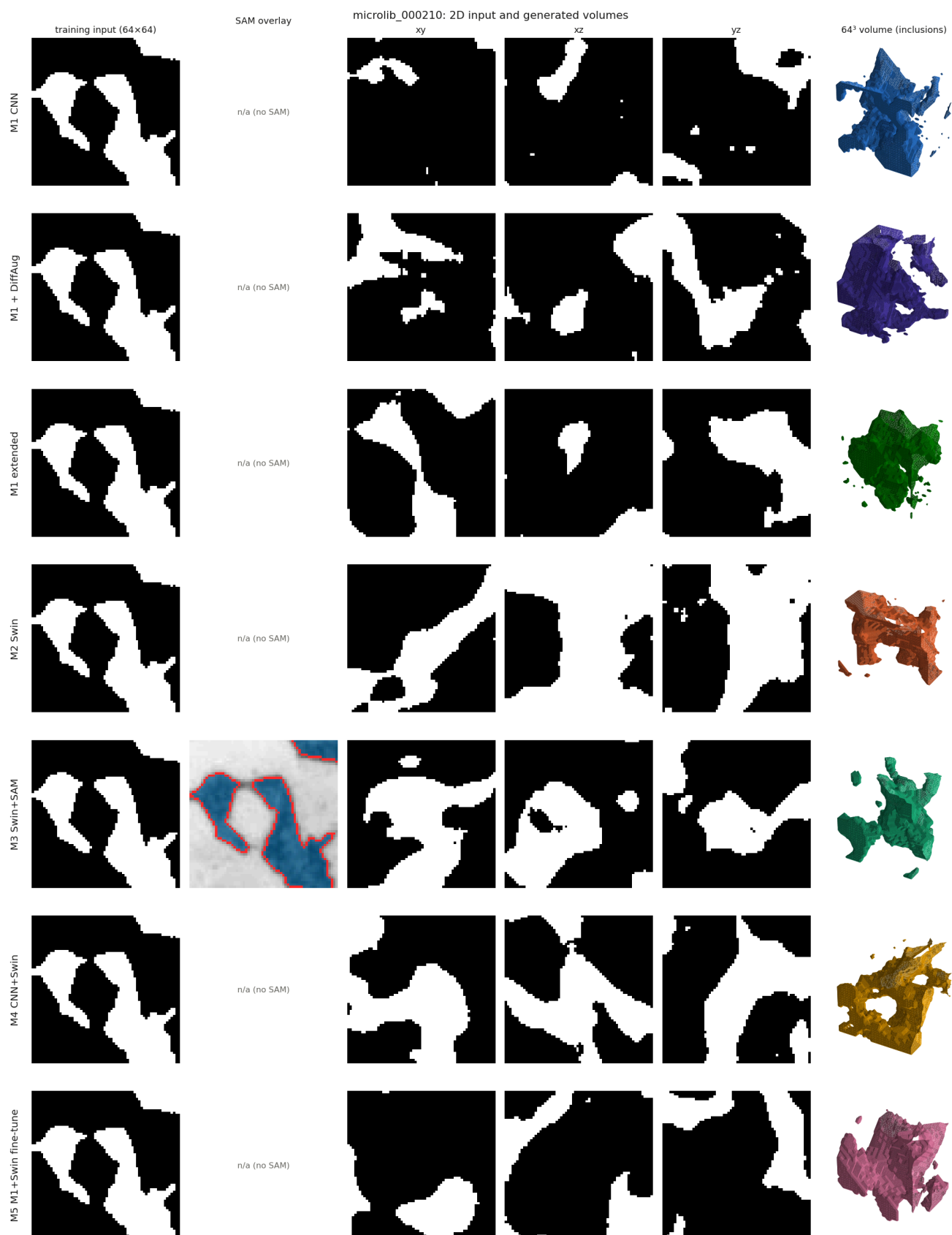
Model	Checkpoint	ϕ	$ \Delta\phi $	S_2 MAE	L MAE	Typical S_2	vs M1
M1 CNN (SliceGAN)	last	0.235	0.003 [0.000, 0.012]	0.0029 [0.0008, 0.0087]	0.0011 [0.0003, 0.0059]	0.0051	—
M1 + DiffAug	last	0.229	0.003 [0.000, 0.012]	0.0009 [0.0006, 0.0066]	0.0014 [0.0010, 0.0059]	0.0136	no clear difference
M2 Swin + DiffAug	best	0.239	0.007 [0.000, 0.017]	0.0077 [0.0017, 0.0141]	0.0055 [0.0010, 0.0109]	0.0132	no clear difference
M3 Swin + DiffAug, SAM map	last	0.174	0.058 [0.050, 0.066]	0.0328 [0.0281, 0.0371]	0.0262 [0.0223, 0.0299]	0.0143	worse (all metrics)
M4 CNN + frozen Swin	best (= last)	0.232	0.000 [0.000, 0.009]	0.0014 [0.0006, 0.0067]	0.0015 [0.0006, 0.0058]	0.0096	no clear difference
M5 M1 + Swin fine- tune	last	0.243	0.010 [0.001, 0.020]	0.0097 [0.0033, 0.0161]	0.0092 [0.0038, 0.0147]	0.0072	worse on L
M1 extended	last	0.238	0.006 [0.000, 0.015]	0.0059 [0.0010, 0.0124]	0.0059 [0.0019, 0.0114]	0.0097	no clear difference

All errors are far below the random-volume floor (S_2 MAE 0.042, L MAE 0.077). The CNN-based models (M1, M1 + DiffAug, M4) form the best group: their point estimates are the lowest and their intervals overlap. In this run M4, the CNN + frozen-Swin ensemble, reproduces the phase fraction exactly (0.232) and needed no checkpoint selection (its best epoch is its last); a second run (Section 5.7) does not repeat either property. No model is significantly better than M1 with 128 test volumes; M3 is significantly worse, because its training image (the SAM map) has a lower phase fraction and different morphology than the Otsu reference, and its last checkpoint undershoots ϕ further.

microlib_000210: generated 64^3 volumes (mean over slices and seeds) vs 2D Otsu label map



Descriptor curves



Qualitative comparison

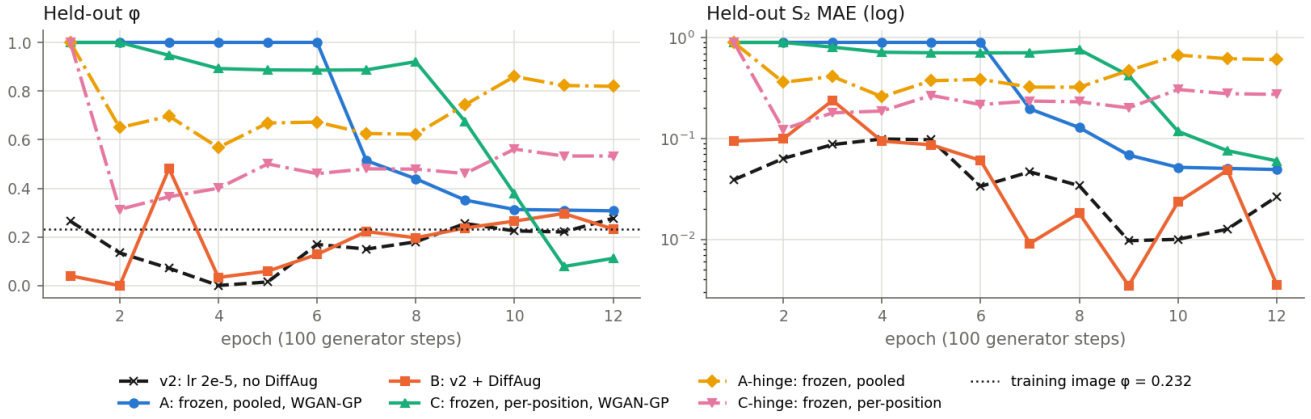
Per-model training curves: [figures/microlib_000210_training.png](#); all volumes can be explored in the interactive viewer ([reports/viewer/](#)).

5.2 Training a Swin critic: stabilization study

Transformer critics are known to make GAN training unstable (ViTGAN, Lee et al., 2022). With SliceGAN’s WGAN-GP setup the Swin-T critic was unstable in every configuration tried; the table summarizes the first 12 epochs on MicroLib (held-out φ target 0.232; S_2 MAE of M1’s best epochs: 0.002–0.004).

Run	Swin critic	Loss	DiffAug	Epochs 1–12	Outcome
v1	stages 3–4 trainable, lr 1e-4, last checkpoint	WGAN-GP	no	realistic at epoch 10, empty at 15	oscillates, ended collapsed ($\varphi = 0.003$)
v2	stages 3–4, lr 2e-5	WGAN-GP	no	empty at 4–5, S_2 MAE 0.010 at 9–11	oscillates, M1-level epochs exist
v2 rerun	identical settings and seed	WGAN-GP	no	good at epoch 2, then $\varphi \approx 0$ for 10 epochs	run-to-run variability: non-deterministic GPU kernels, amplified by the GAN dynamics
v3	v2 + ViTGAN: improved spectral norm, Adam $\beta_1 = 0$, G EMA	WGAN-GP	no	φ 0.002–0.08 throughout	critic too strong
stage-4 probe	only stage 4 trainable	WGAN-GP	no	φ 0.08–0.29, best S_2 MAE 0.037	no full collapse, blurry
A	frozen, pooled multi-scale head	WGAN-GP	yes	$\varphi = 1.0$ at epochs 1–6, then $0.51 \rightarrow 0.31$; best S_2 MAE 0.049	fails: gradient penalty cannot be met through a frozen backbone
B	stages 3–4, lr 2e-5	WGAN-GP	yes	$\varphi \approx 0$ at epochs 1–5; from epoch 7 φ 0.20–0.30, S_2 MAE 0.0091 / 0.018 / 0.0035 / 0.024 / 0.049 / 0.0035	passes: M1-level epochs, more consistent than v2
C	frozen, per-position multi-scale heads	WGAN-GP	yes	$\varphi = 1.0$ at epochs 1–2, 0.89–0.95 to epoch 8, then $0.38 \rightarrow 0.08 \rightarrow 0.11$; best S_2 MAE 0.060	fails like A: the GP, not the pooling, is the problem
A-hinge	frozen, pooled head, head spectral norm	hinge	yes	φ 0.57–1.0, drifting up; best S_2 MAE 0.26	fails: no collapse, but φ uncontrolled
C-hinge	frozen, per-position heads, head spectral norm	hinge	yes	φ 0.31–0.56; best S_2 MAE 0.12	fails alone; better than A-hinge, used inside M4/M5

Swin critic probes on MicroLib 000210: held-out ϕ and S_2 MAE per epoch (first 12 epochs)



Swin critic probes

Per-epoch logs of every probe are in `reports/probes/`.

Decision. M2/M3 use probe B (the only Swin-only critic reaching M1-level epochs); M4/M5 use the per-position frozen heads of C-hinge inside an ensemble with the CNN critic.

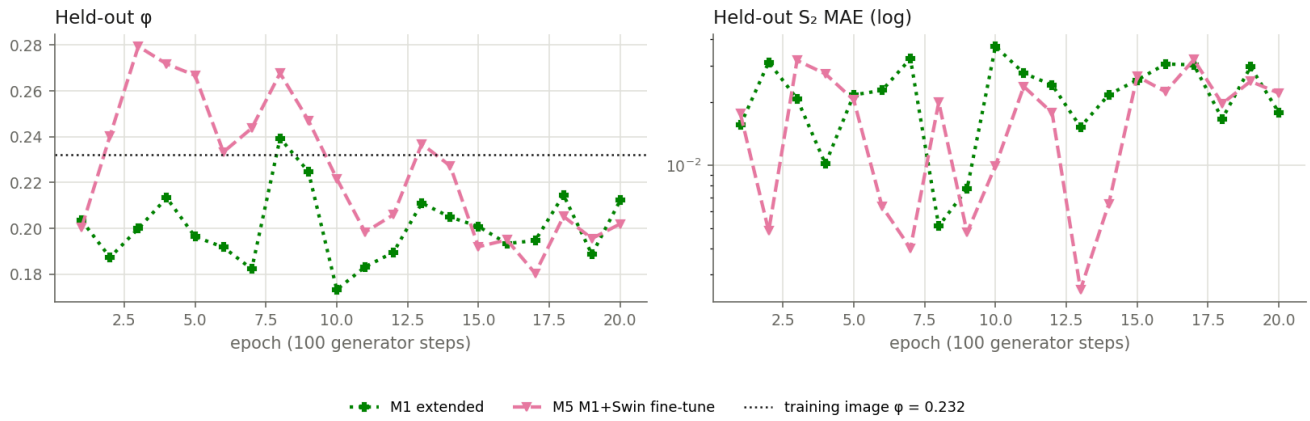
Three findings shaped the final models. First, a frozen pretrained backbone cannot be trained with WGAN-GP: the gradient penalty asks for unit input-gradient norm, which the frozen Swin layers fix and the small head can only rescale, so the penalty dominates (values of 10–355 instead of ≈ 1) and the critic gives no useful signal. Frozen-backbone critics in the literature (Projected GAN, Vision-aided GAN) use hinge or BCE losses instead. Second, Vision-aided GAN reports that pretrained critics used *alone* diverge and help only in an ensemble with the original discriminator, which motivates M4 and M5. Third, with a hinge loss the frozen Swin critic no longer collapses but does not control the phase fraction on its own (ϕ drifts to 0.3–0.8); a plausible cause is Swin’s per-token LayerNorm, which removes much of the absolute phase-intensity information. The CNN critic in M4/M5 supplies that constraint. Finally, DiffAug is what made the trainable Swin critic work (B vs v2), in line with the limited-data GAN literature (Zhao et al., 2020; Karras et al., 2020): the critic otherwise overfits the few hundred distinct views of a single micrograph.

Linear-probe diagnostic (Vision-aided GAN, Sec. 3.2): a logistic regression on frozen, pooled Swin features separates real 64×64 crops from slices of M1’s best generator with 73 % held-out accuracy at 64 px input (stage 2 alone: 75 %), 83 % at 128 px and 83 % at 224 px; random volumes with the correct ϕ are separated at 100 %. The frozen features therefore carry a usable real-vs-fake signal, strongest at the 8×8 resolution of stage 2, and a larger input could add about 10 points.

5.3 Fine-tuning with a Swin critic (M5 vs M1-extended)

Both runs start from M1’s selected checkpoint and train 20 more epochs with fresh critics; they differ only in the critics (CNN vs CNN + frozen Swin with per-position heads). Neither improves on M1: on the test volumes M1-extended reaches S_2 MAE 0.0059 and M5 0.0097 (M1: 0.0029), and their best per-epoch held-out scores were reached in the first epoch, i.e. at the starting point. Once SliceGAN has converged, adding the Swin critic as a fine-tuning stage (the Vision-aided GAN recipe) does not help here. An earlier version of this comparison (run v5, selection on 2 seeds) suggested the opposite; that result was selection noise and disappeared on more seeds (Section 3). On the synthetic dataset (Section 5.6) both runs improve on M1 and M5 has the lowest S_2 MAE of all models (0.0013 vs 0.0020 for M1-extended), but the difference between them is not significant (95 % interval of M5 – M1-extended: -0.0028 to $+0.0023$): the gain comes from the extra training, not from the Swin critic.

Fine-tuning M1's best generator for 20 epochs: CNN only (M1 extended) vs CNN + Swin (M5)



M5 vs M1 extended (run v5 selection traces)

5.4 SAM front-end

Dataset	ϕ SAM	ϕ Otsu	Reference (ϕ)	SAM IoU / Dice	Otsu IoU / Dice
synthetic	0.290	0.254	exact ground truth (0.250)	0.864 / 0.927	0.913 / 0.955
MicroLib 000210	0.217	0.232	MicroLib-annotated threshold (0.219)	0.845 / 0.916	0.941 / 0.970

Zero-shot SAM needs tiling on this image (a single pass with 16 points per side finds $\phi = 0.07$), and even tiled it has lower IoU/Dice than Otsu on both references. The EDA notebook ([notebooks/00_eda.ipynb](#)) shows where the errors come from:

- **Synthetic (true ground truth):** SAM's error is a systematic one-pixel dilation. All of its wrong pixels lie within 2 px of a true interface and almost all are false inclusion, which adds up to ϕ overestimated by 16 %. Otsu's errors sit at the same interfaces but are balanced, so its ϕ is almost exact.
- **MicroLib:** the two maps agree on 96 % of the pixels. Two thirds of the disagreement is the gray halo around each dark particle, which Otsu (threshold 79, above the curated midpoint 70) labels as inclusion. The rest is a few mid-gray regions that SAM takes as whole objects. As a result SAM's ϕ , S_2 and L are *closer* to the curated reference than Otsu's (S_2 MAE 0.0009 vs 0.0085). Otsu's IoU is still higher, because the reference is itself a global threshold and structurally favours Otsu.

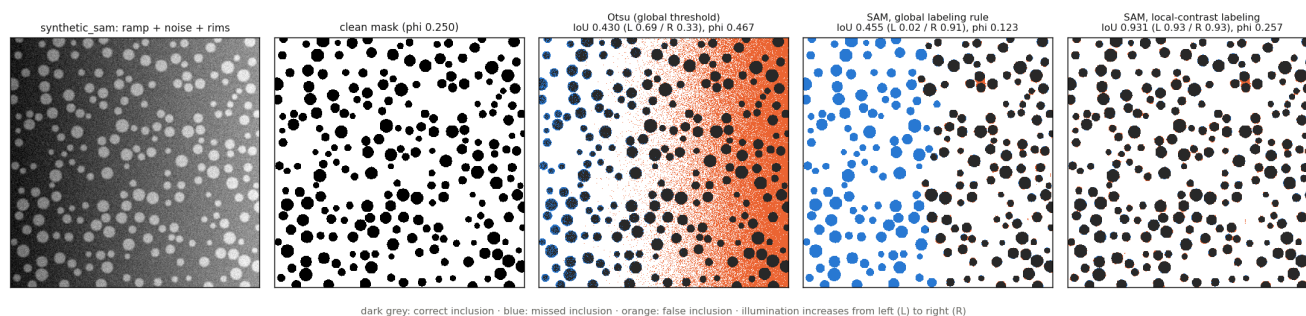
The choice of front-end therefore moves the GAN's *target* by about 0.009 in S_2 MAE, more than the gap between the best MicroLib models (0.001–0.003). This is why every model is scored against its own training map and, separately, against a common reference (§4).

The EDA also shows that the micrograph is mildly banded along x: the correlation length is about 23 px along x vs 19 px along y (S_2 within 5 % of its plateau). SliceGAN assumes the same statistics on all three planes, and the D4 crop augmentation symmetrizes x and y on purpose, so the generators learn an isotropic version of the structure. The radially averaged metrics are insensitive to this, but no model here reproduces the banding direction.

Label probe: an image where a global threshold fails. Both training images above favour a global threshold, so they cannot show what SAM adds. A third synthetic image, `synthetic_sam` (`configs/data/synthetic_sam.yaml`), keeps the same non-overlapping discs ($\phi = 0.250$, own layout) but adds an illumination ramp across x larger than the phase contrast, per-phase noise that makes the two gray histograms overlap, and a thin dark rim around each disc so every object stays locally visible. It is a label probe,

not a training set: Otsu and SAM are scored against its clean mask (`src/features/sam_probe.py`, `reports/sam_probe.json`), and no GAN is trained on it.

Labeling (vs clean mask, ϕ 0.250)	IoU	IoU dark half / bright half	ϕ
Otsu (global threshold)	0.430	0.691 / 0.329	0.467
SAM, global labeling rule (as used for M3)	0.455	0.019 / 0.909	0.123
SAM, local-contrast labeling	0.931	0.932 / 0.930	0.257

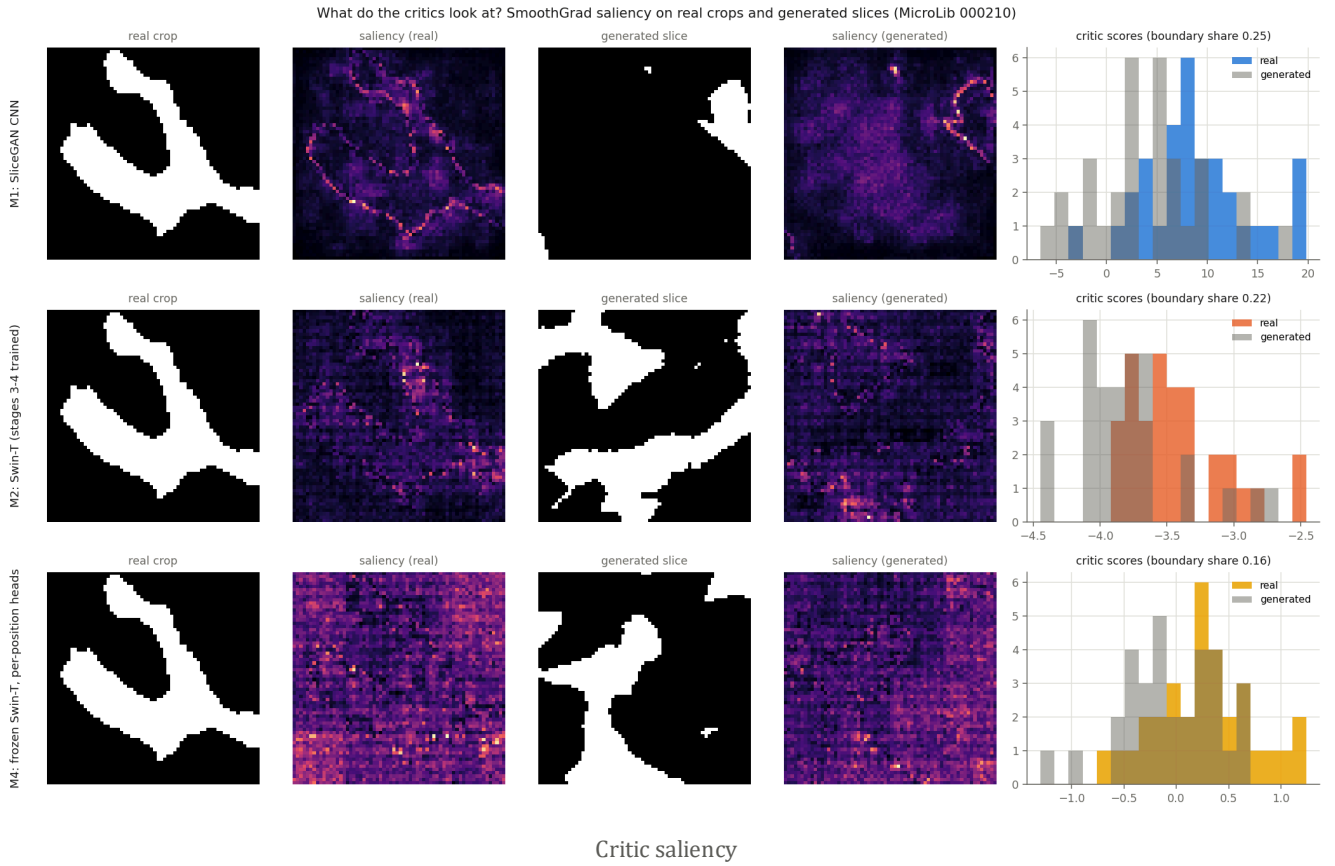


SAM label probe

Otsu cuts across the ramp: it misses inclusions on the dark side and labels most of the bright-side matrix as inclusion (ϕ 0.467). SAM's masks find every disc, but the pipeline's rule that assigns a phase to each mask group is itself global (one split of the groups' mean gray levels), so all discs on the dark half, which are darker than the bright-half matrix, are labelled matrix. Scoring each group against a 3 px ring just outside it instead (`sam.classify: local`, opt-in; M3 used the global rule) cancels the ramp: IoU 0.931 and ϕ within 0.007 of the truth, above Otsu on the clean synthetic image (0.913). SAM's instance masks are therefore not the weak point; the step that turns masks into phases is.

5.5 What do the critics look at?

SmoothGrad saliency (Smilkov et al., 2017): the gradient of each trained critic's score with respect to its one-hot input, averaged over 16 noisy copies, on 32 real crops and 32 generated slices (`src/visualization/critic_saliency.py`, values in `reports/critic_saliency.json`).



Critic	Saliency on phase boundaries (boundary band = 12 % of pixels)	P(score real > score generated)
M1 SliceGAN CNN	25 % (2.1 × its area)	0.75
M2 Swin-T, stages 3–4 trained	22 % (1.8 ×)	0.79
M4 frozen Swin-T, per- position heads	16 % (1.3 ×)	0.67

The CNN critic concentrates on the inclusion outlines: it judges interfaces. The trained Swin-T also favours boundaries but more diffusely, and its saliency shows the 4×4 patch grid of the transformer. The frozen Swin-T heads spread their attention over the whole slice: they respond to global texture and arrangement rather than to local edges, and separate real from generated slices least on their own. The two critics of M4 therefore use complementary cues (CNN: interfaces; frozen Swin: global texture), a plausible reason why the ensemble trains reliably, while the frozen Swin alone cannot control the phase fraction (Section 5.2).

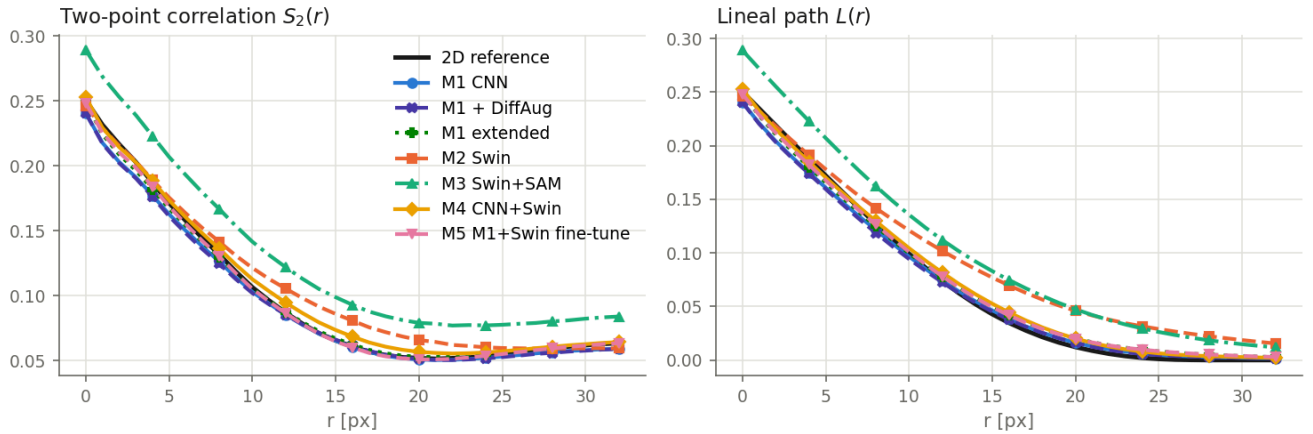
5.6 Synthetic dataset: comparison against a true ground truth

All seven models were retrained on the synthetic micrograph with the same configurations and protocol. Here the common reference is the exact ground-truth mask ($\varphi = 0.250$), so the scores measure how close each model

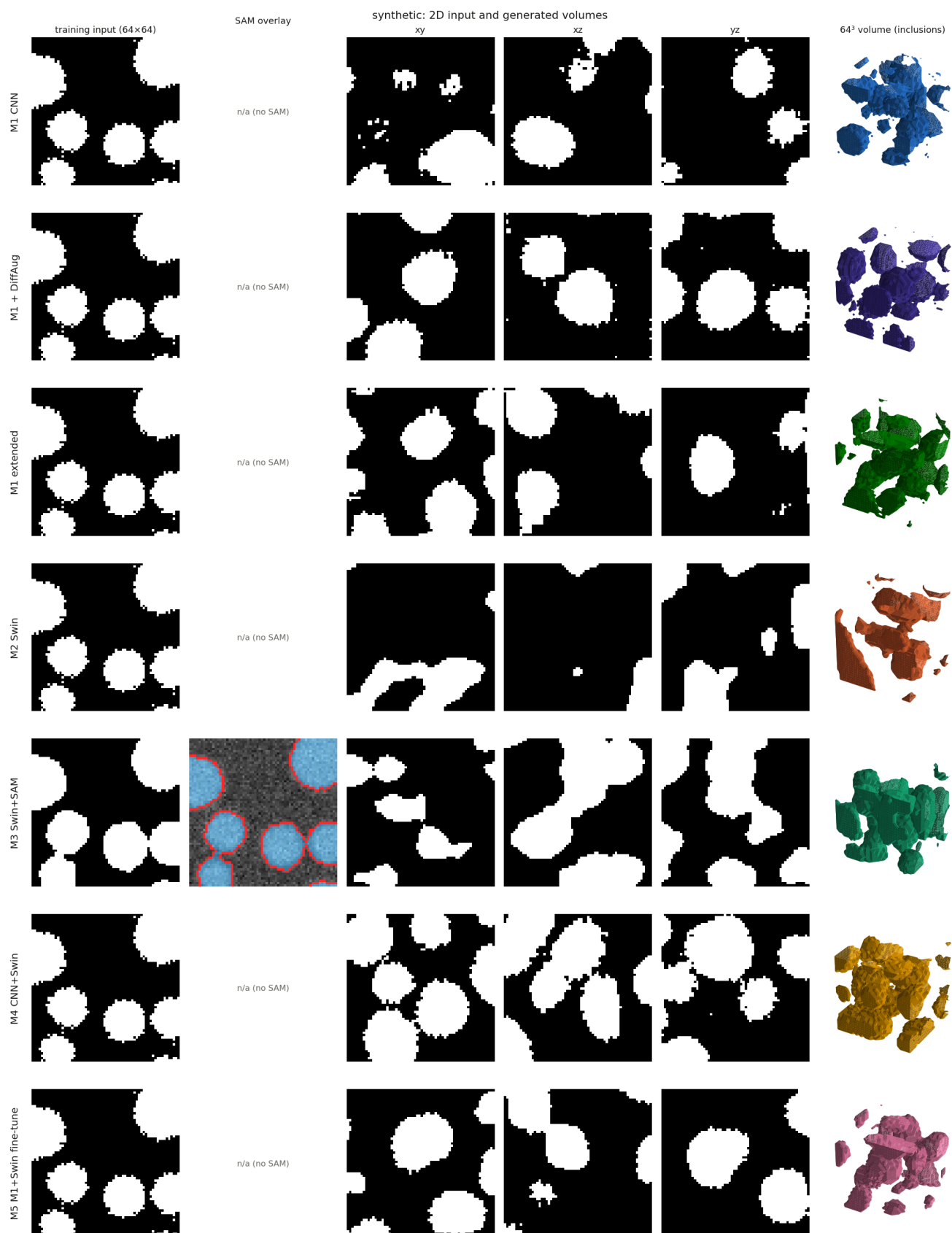
gets to the *true* structure, including the error of its segmentation front-end. For scale, the Otsu map itself scores S_2 MAE 0.0021 against this mask.

Model	Checkpoint	ϕ	$ \Delta\phi $	S_2 MAE	L MAE	Typical S_2	vs M1
M1 CNN (SliceGAN)	snapshot ep. 30	0.240	0.010 [0.003, 0.017]	0.0040 [0.0023, 0.0080]	0.0052 [0.0042, 0.0066]	0.0084	—
M1 + DiffAug	snapshot ep. 20	0.240	0.010 [0.004, 0.017]	0.0045 [0.0028, 0.0080]	0.0056 [0.0043, 0.0072]	0.0074	no clear difference
M2 Swin + DiffAug	best (ep. 29)	0.246	0.004 [0.000, 0.013]	0.0093 [0.0075, 0.0140]	0.0216 [0.0189, 0.0260]	0.0150	worse (S_2 , L)
M3 Swin + DiffAug, SAM map	best (ep. 15)	0.289	0.039 [0.032, 0.045]	0.0298 [0.0252, 0.0344]	0.0317 [0.0282, 0.0350]	0.0106	worse (all metrics)
M4 CNN + frozen Swin	snapshot ep. 35	0.253	0.002 [0.000, 0.008]	0.0039 [0.0026, 0.0075]	0.0052 [0.0046, 0.0067]	0.0032	no clear difference
M5 M1 + Swin fine- tune	best (ep. 8 of 20)	0.248	0.002 [0.000, 0.008]	0.0013 [0.0008, 0.0046]	0.0053 [0.0049, 0.0061]	0.0050	no clear difference
M1 extended	best (ep. 8 of 20)	0.249	0.002 [0.000, 0.008]	0.0020 [0.0018, 0.0045]	0.0056 [0.0051, 0.0065]	0.0050	no clear difference

synthetic: generated 64^3 volumes (mean over slices and seeds) vs 2D ground-truth mask



Synthetic descriptor curves



Synthetic qualitative comparison

The synthetic results confirm the MicroLib conclusions (one training run per model; Section 5.7 shows how much a second run can differ):

- **Swin as the only critic (RQ1).** M2 is significantly worse than M1 on S_2 and L, and also worse than the fair ablation M1 + DiffAug (S_2 difference +0.0048 [+0.0006, +0.0098], L +0.0161 [+0.0129, +0.0206]). Its volumes lose the disc/sphere morphology (elongated, merged inclusions in the qualitative panel). They are also **anisotropic**: xy slices match well (S_2 MAE 0.0054), xz and yz slices do not (0.0134 and 0.0101). The first MicroLib run of M2 shows the same pattern (xy 0.0027 vs xz 0.0114 and yz 0.0098), but its second MicroLib run is isotropic (Section 5.7), so the anisotropy is a failure mode of some Swin-critic runs, not a systematic property; the CNN-based models are isotropic in every run.
- **SAM front-end (RQ2).** M3 reproduces its own training map faithfully (S_2 MAE 0.0029 vs the SAM map, ϕ 0.289 vs 0.290), so the GAN works; the error comes entirely from the front-end, whose one-pixel dilation (Section 5.4) the generator learns as a real feature. Against the true structure M3 is the worst model and significantly worse than M2, its Otsu-trained counterpart, on every metric.
- **Swin as an additional critic (RQ3).** M4 is level with M1 on every metric, reproduces ϕ within 0.002, is significantly better than M2 (S_2 -0.0054 [-0.0102, -0.0015], L -0.0163 [-0.0207, -0.0134]), and was the most stable model during training in this run: its selection-free median held-out S_2 MAE (0.0032, interquartile range 0.0023–0.0039) is the lowest of all seven, less than half of M1’s (0.0084). As a fine-tuning stage, M5 gives the lowest test S_2 MAE (0.0013, below the Otsu segmentation’s own 0.0021, because the generator smooths out the segmentation noise), but it is not distinguishable from M1-extended.

5.7 Repeat runs and a Swin critic at 128 px (MicroLib)

Two follow-up experiments test how robust the conclusions are. First, M1, M2 and M4 were trained a second time with a different training seed (43 instead of 42) and otherwise identical configuration; the validation and test seeds are the same as before. Second, M4 was trained with its Swin branch at 128 px: slices are upsampled from 64 to 128 px before the frozen Swin-T, so its attention windows stay 7×7 in the first stages, motivated by the linear probe (73 % at 64 px vs 83 % at 128 px). This run needed one generator backward pass per slice orientation instead of one for all three (the same gradient, a third of the critic activations in memory); otherwise it ran out of GPU memory. Values: `reports/metrics_seeds_microlib.csv`.

Model	Run	Checkpoint	ϕ	$ \Delta\phi $	S_2 MAE	L MAE	Typical S_2	S_2 MAE xy / xz / yz
M1 CNN	1	last	0.235	0.003 [0.000, 0.012]	0.0029 [0.0008, 0.0088]	0.0011 [0.0003, 0.0059]	0.0051	0.0047 / 0.0008 / 0.0040
M1 CNN	2	snapshot ep. 35	0.220	0.012 [0.003, 0.020]	0.0055 [0.0012, 0.0112]	0.0045 [0.0015, 0.0093]	0.0127	0.0074 / 0.0048 / 0.0042
M2 Swin + DiffAug	1	best	0.239	0.007 [0.000, 0.017]	0.0077 [0.0019, 0.0140]	0.0055 [0.0009, 0.0108]	0.0132	0.0027 / 0.0114 / 0.0098
M2 Swin + DiffAug	2	last	0.229	0.003 [0.000, 0.012]	0.0017 [0.0007, 0.0072]	0.0010 [0.0004, 0.0060]	0.0067	0.0051 / 0.0043 / 0.0049
M4 CNN + frozen Swin	1	best (= last)	0.232	0.000 [0.000, 0.009]	0.0014 [0.0006, 0.0068]	0.0015 [0.0006, 0.0059]	0.0096	0.0019 / 0.0075 / 0.0014
M4 CNN + frozen Swin	2	snapshot ep. 35	0.222	0.010 [0.002, 0.018]	0.0044 [0.0015, 0.0096]	0.0043 [0.0012, 0.0087]	0.0104	0.0065 / 0.0027 / 0.0060
M4, Swin at 128 px	1	snapshot ep. 40	0.240	0.008 [0.000, 0.016]	0.0042 [0.0013, 0.0100]	0.0022 [0.0007, 0.0070]	0.0084	0.0065 / 0.0014 / 0.0056

- **Run-to-run variability is as large as the differences between models.** For each model the two runs are not significantly different, but their point estimates differ by factors of 2–4 (M2: S_2 MAE 0.0077 vs 0.0017). The ranking of M1, M2 and M4 changes between the first and the second set of runs, and within the second set no pair is significantly different. On MicroLib, the three critics therefore reach the same quality; a ranking from a single run per model would not be reliable.
- **Properties that did not repeat.** M4’s exact ϕ and “best epoch = last”, and M2’s anisotropy, were features of single runs. M4’s selection-free typical score is the most reproducible between runs (0.0096 and 0.0104, vs 0.0051 and 0.0127 for M1), which is weak evidence for a more predictable training, not for a better one.
- **Swin at 128 px.** No clear difference from M4 at 64 px (S_2 difference +0.0018 [−0.0037, +0.0086]), M1 or M2. Its selected checkpoint scored 0.0010 on the validation seeds but 0.0042 on the test seeds, and its late epochs oscillate (last epoch 0.027). The better real-vs-generated separation of the frozen features at 128 px did not translate into better volumes, so the synthetic repeat of this variant was not run.

5.8 Model size and computational cost

Measured on the RTX A2000 (12 GB) used for every run (`src/visualization/compute_cost.py`, `reports/compute_cost.json`). FLOPs count 2 per multiply-add on one input; the training estimate per generator step follows SliceGAN’s schedule (the critic sees 1464 slices and the generator 7 volumes per step, forward plus backward $\approx 3\times$ forward; gradient penalty and DiffAug not included).

Component	Parameters (trainable)	GFLOPs per forward pass	Used for
3D generator (all models)	40.1 M	27.1 per 64^3 volume (145 per 128^3)	training and inference
SliceGAN CNN critic	2.76 M	0.21 per 64×64 slice	training (M1, M4, M5)
Swin-T critic, stages 3–4 trained	27.5 M (26.3 M)	0.85 per slice	training (M2, M3)
Frozen Swin-T + per-position heads	27.7 M (0.18 M)	0.85 per slice (4.1 at 128 px)	training (M4, M5)
SAM ViT-B (front-end)	93.7 M (none trained)	—	M3 data preparation, once

Model	Estimated TFLOPs per generator step	Measured s per generator step	Training time (MicroLib)
M1 CNN	1.5	0.31	27 min
M1 + DiffAug	1.5	0.40	36 min
M2 / M3 Swin critic	4.3	1.93	163 / 162 min
M4 CNN + frozen Swin	5.2	1.16	98 min
M4, Swin at 128 px	19.5	3.61	303 min
M5 / M1-extended (20 epochs from M1)	5.2 / 1.5	1.16 / 0.31	40 / 11 min (+ M1)

- **Inference costs the same for every model.** The critics are only needed for training; all models share the same 40.1 M-parameter generator, which produces a 64^3 volume in 14 ms on the GPU (0.25 s on CPU) and a 128^3 volume in 67 ms. Vision Transformer critics therefore add no deployment cost.
- **Training cost is set by the critic.** A Swin-T slice costs about $4\times$ the FLOPs of a CNN slice, and the critic is evaluated on about 1500 slices per generator step, so the critic dominates training. M2 / M3 take $6\times$ longer than M1, more than the FLOP ratio ($2.9\times$), because gradients flow through 26 M critic weights and the gradient penalty’s double backward passes through attention. M4 does more FLOPs than M2 but trains faster ($3.7\times$ M1): its Swin backbone is frozen (no weight gradients) and its Swin branch uses a hinge loss without a gradient penalty. Moving the Swin to 128 px multiplies its FLOPs by 4.8 and its training time by 3, for no gain (Section 5.7); it also needed one generator backward pass per slice orientation to fit in 12 GB.
- **SAM is a one-off cost.** Segmenting the MicroLib image takes about 35 s (93.7 M parameters, 256 px tiles), negligible next to training, and SAM is not used at generation time.

- **Cost against benefit.** Since no Swin variant beats the CNN critic (Sections 5.1, 5.6, 5.7), SliceGAN’s 2.8 M-parameter critic is also the best choice per GPU-hour. The final runs of this report took 28 GPU-hours in total; the exploratory runs (v1–v5, probes) are not included.

6. Conclusions and future work

Answers to the project questions (two datasets, 128 test volumes per model, bootstrap intervals; two training runs of M1, M2 and M4 on MicroLib):

- **RQ1 — Swin-T instead of the CNN critic: no.** Trained in SliceGAN’s WGAN-GP setup, a Swin-T critic was unstable in every configuration until DiffAug was added. With DiffAug it reaches the CNN’s quality but not more: on MicroLib its two runs bracket the CNN’s (no significant difference in either), and the single synthetic run is significantly worse on S_2 and L (also against the M1 + DiffAug ablation) and less isotropic. Standard ViT-GAN stabilizers (lower learning rate, improved spectral norm, Adam $\beta_1 = 0$, generator EMA, a smaller trainable part) did not remove the instability; a frozen Swin critic cannot be trained with WGAN-GP at all (the gradient penalty dominates) and, with a hinge loss, does not control φ .
- **RQ2 — SAM instead of a global threshold: no, for this kind of image.** On high-contrast two-phase micrographs zero-shot SAM is a worse segmentation than Otsu against a true ground truth (a systematic one-pixel dilation, $\varphi +16\%$). The GAN reproduces SAM’s map faithfully, so M3 inherits this bias and is significantly worse than M2 on both datasets. The front-end shifts the GAN’s target more than any change of critic does: segmentation quality matters more than the critic architecture. A label probe where a global threshold fails (an illumination ramp, Section 5.4) shows the other side: SAM’s masks still find every particle, and with a local-contrast labeling rule SAM reaches IoU 0.93 against 0.43 for Otsu. In this pipeline the weakness of zero-shot SAM is the step that assigns phases to its masks, not the masks.
- **RQ3 — Swin-T as an additional critic: it matches the baseline, with no measurable gain.** The CNN + frozen-Swin ensemble (M4, Vision-aided GAN style) is statistically level with SliceGAN in all three comparisons (two MicroLib runs, one synthetic), is significantly better than the Swin-only critic on synthetic data, and its training quality is the most reproducible between runs. Its first run looked better (exact φ , best epoch = last, lowest typical score on synthetic), but the second MicroLib run did not repeat that. Saliency maps show that the two critics use complementary cues: the CNN judges interfaces, the frozen Swin global texture. Moving the Swin branch to 128 px did not help. Used as a fine-tuning stage of a converged SliceGAN (M5), the Swin critic gives no gain over training the CNN alone for the same extra steps.

Overall. SliceGAN’s small CNN critic is hard to beat on two-phase 64^3 microstructures. A Swin-T critic, alone (with DiffAug) or next to the CNN, reaches the same quality but not better; used alone it is harder to train and sometimes produces anisotropic volumes, while next to the CNN (a 0.18 M-parameter head on a frozen backbone) it trains as reliably as the baseline. The second lesson is methodological and probably the most transferable: single 64^3 volumes are noisy samples ($\varphi \pm 0.05$), the same model trained twice can differ by a factor of 2–4, and model rankings obtained with a few seeds or a single run reversed when re-evaluated. Separate selection, validation and test seeds, bootstrap intervals and repeat runs were needed to reach conclusions that hold.

Limitations. Two training runs for M1, M2 and M4 on MicroLib and one for every other model and for the synthetic dataset, while Section 5.7 shows that run-to-run variability is as large as the differences between models; 64 px slices, which force Swin-T’s windows down to 4×4 and 2×2 and use the backbone far from its 224 px pretraining resolution; one real micrograph, high contrast and two phases, where a global threshold is already near-optimal.

SliceGAN is the literature baseline; 2024 works (Micro3Diff, Lee & Yun 2024; DDPM-GAN, Phan et al. 2024) improve descriptors and stability with diffusion models, but are outside the scope of this work. The contribution here is a controlled evaluation of Swin critics and of SAM as a phase front-end.

Ongoing work: local-contrast labeling on the training images. The labeling rule that wins the probe (Section 5.4) is being applied to the existing synthetic and MicroLib images, writing to separate folders so the maps M3 was trained on are unchanged, and scored against the same references as Otsu and the global rule. On MicroLib it could remove the gray halo that separates SAM’s map from Otsu’s; if the new maps are better, M3 will be retrained on them. Results are not part of this report.

Future work. Five or more training runs per model, which the run-to-run spread requires before any ranking; harder micrographs (low contrast, texture, three phases) where SAM’s object-level segmentation can pay off, possibly with point prompts or a fine-tuned mask decoder; and diffusion-based generators.

7. Planning

Task	Owner	Status
Repo skeleton, configs, Makefile, Poetry setup, tests, MLflow tracking	Braian Desia	Done
Datasets: synthetic + MicroLib 000210; EDA notebook	Braian Desia	Done
Descriptors φ / S_2 / L , bootstrap intervals + tests	Braian Desia	Done
M1 SliceGAN baseline (+ DiffAug ablation, M1-extended)	Braian Desia	Done
M2 Swin-T critic + stabilization study (v1–v3, probes A/B/C, hinge variants)	Braian Desia	Done
M3 SAM front-end	Braian Desia	Done
M4 / M5 Vision-aided extensions	Braian Desia	Done
Repeat runs (second seeds of M1, M2, M4) and Swin critic at 128 px	Braian Desia	Done
Evaluation protocol (train / validation / test seeds), figures, saliency	Braian Desia	Done
Interactive volume viewer and Streamlit explorer	Braian Desia	Done
RVE export for FEM / FFT codes	Braian Desia	Done
SAM label probe (<code>synthetic_sam</code>) and local-contrast labeling	Braian Desia	Done
Local-contrast labeling on the synthetic and MicroLib images (ongoing work, Section 6)	Braian Desia	In progress
Report (English + Spanish, PDF)	Braian Desia	Done (5 Oct)
Presentation, 15 min	Braian Desia	12 Oct

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